
ErSE 222 - Machine Learning in Geoscience: Midterm Exam

Duration: 1 h

Q1: Logistic Regression can be implemented using a simple single perceptron.

- Give a detailed illustration of the main components of Logistic Regression, and draw a simple schematic.
- How will you update the weight and bias of the single perceptron (estimate the gradient)?
- What does it mean to replace the last layer of the Logistic Regression with a linear activation function?

Q2: In your research team, you are working on classifying different rock types based on some physical properties. You have 2 rock types, and each has 10 physical properties. According to that, answer the following questions:

- Your first trial will be using a 3-layer MLP network shown in Figure 1. What will be your loss function? Estimate the gradient for all the weights and biases in the network.
- What are the evaluation metrics you will use to assess the performance of your network?
- What will you change in your network and loss function if the types of rocks increase to 5?

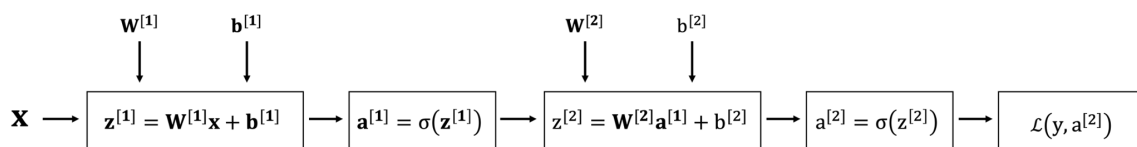


Figure 1: 3-layer MLP network.

Q3: You belong to a seismic interpretation research group, and you are aiming to enhance seismic imaging. The resolution of seismic images often suffers due to inadequate survey equipment and the complexity of subsurface structures. This degradation in resolution can make it challenging to accurately interpret the subsurface structures. Thus, the main objective is seismic image enhancement, meaning your input data is the low-resolution (low frequency) noisy seismic images, and your target is the high-resolution (high-frequency) seismic images.

- Give a general concept on how to design a robust machine learning model to fit with the problem, specifying your steps.
- What network will you choose? Give a detailed explanation of the network with the role of each layer. You can use a schematic if needed.
- How will you divide the data, and what loss function will you use?
- What will be the number of input and output channels in this case?
- How will you evaluate your results?

Q4: Answer the following questions:

- What are the limitations of the MLP?
- How does PINN differ from the MLP and what applications can PINN fit?
- What are the advantages of using CNN over MLP?
- What are the limitations of using Gradient Descent as an optimizer for updating the parameters of neural networks, and how can these problems be tackled?
- The Adam optimizer combines two concepts; illustrate how it works.
- What are the advantages of using activation functions in neural networks?
- What is the meaning of gradient vanishing?